DELHI TECHNOLOGICAL UNIVERSITY

(Formerly Delhi College of Engineering)

Bawana Road, Delhi – 110042

B.TECH PROJECT REPORT

Submitted in partial fulfilment of the requirements for the degree of

Bachelor of Technology in Computer Science and Engineering

Can an AI Explanation Be Misused?

Testing Whether SHAP and LIME Help Patients Game a Clinical Diabetes Risk Model

---

Submitted by:

Akash Singh	Roll No.: 23/CS/036
Aditya Jha	Roll No.: 23/CS/025
Akshat Kumar	Roll No.: 23/CS/039

Under the supervision of:

Miss. Anjali Bansal
Assistant Professor, Department of CSE

Department of Computer Science and Engineering

Academic Year 2026–2027

Certificate

This is to certify that the B.Tech Project Report titled **"Can an AI Explanation Be Misused? Testing Whether SHAP and LIME Help Patients Game a Clinical Diabetes Risk Model"** submitted by **Akash Singh** (Roll No. 23/CS/036), **Aditya Jha** (Roll No. 23/CS/025), and **Akshat Kumar** (Roll No. 23/CS/039) in partial fulfilment of the requirements for the award of the degree of *Bachelor of Technology in Computer Science and Engineering* at Delhi Technological University, Delhi, is a record of bonafide work carried out by them under my supervision.

Supervisor:	**Head of Department:**
Miss. Anjali Bansal	[Name of HOD]
Assistant Professor	Professor & Head
Department of CSE, DTU	Department of CSE, DTU
Date: _____	Date: _____

Declaration

We hereby declare that the work presented in this B.Tech Project Report titled "Can an AI Explanation Be Misused?" submitted to the Department of Computer Science and Engineering, Delhi Technological University, is our own original work carried out under the guidance of Miss. Anjali Bansal.

We further declare that:

- The work contained in this report has not been submitted elsewhere for any other degree or diploma.
- Wherever contributions of others have been used, they have been duly acknowledged.
- The analysis plan for the final experiment was recorded before the held-out dataset was examined, as a git tag (`v0.1-preregistration`) in the project repository at https://github.com/Akshat-stack47/BTP-XAI-Gaming. That dataset was later examined before the analysis was corrected; this, and every deviation, is disclosed in Section 3.7.1. The corrected analysis was pre-registered again (`v0.2-preregistration`) before the confirmatory NHANES 2021–2023 data was processed.

Akash Singh _____ Date: _____

Aditya Jha _____ Date: _____

Akshat Kumar _____ Date: _____

Acknowledgements

We express our sincere gratitude to our project supervisor, **Miss. Anjali Bansal**, Assistant Professor, Department of CSE, Delhi Technological University, for her invaluable guidance, constant encouragement, and constructive feedback throughout the course of this project.

We thank the **Centers for Disease Control and Prevention (CDC)** for making the NHANES dataset publicly available, and the developers of the open-source libraries scikit-learn, SHAP, and LIME that made this research possible.

Abstract

Explainable AI (XAI) methods such as SHAP and LIME are increasingly required by regulation to accompany machine learning predictions in healthcare. A concern is that feature attributions may also serve as instructions for gaming the model—telling a patient exactly which self-reported inputs to adjust for a better risk score. However, a determined adversary can also simply query the model repeatedly with different inputs (blind trial-and-error) without any explanation.

We compare these two adversarial strategies on a diabetes risk model trained on NHANES public health survey data. The adversary's permissible lies follow actual human self-reporting errors measured in NHANES—in the direction people really misreport (height up, weight down) and within the 95th percentile of the observed error—making the threat model empirically grounded. We train four models (Random Forest and Logistic Regression, each with nurse-measured or self-reported height/weight inputs) and measure how many blind attempts are required to match the advantage of one explanation: the *explanation equivalence* N^*.

In a pre-registered confirmatory test on the NHANES 2021–2023 cycle, which no one on the team had analysed (1,126 test patients), one SHAP explanation is worth $N^* = 1.42$ blind attempts (95% CI 1.21–1.62) for the primary configuration, and N^* ranges from 0 to 1.42 across all configurations: an explanation is worth about one or two ordinary attempts at the model. An exploratory analysis of the 2017–2018 cycle ($N^* = 0$, upper 95% bound ≤ 1.2) and a comparison on a second survey (BRFSS, 253,680 respondents; $N^* = 0$) agree. For the Random Forest, more than a quarter of explanation-guided lies raise the score instead, while the model that uses nurse-measured inputs cannot be gamed and is as accurate. These results show that **withholding XAI explanations is not an effective safeguard against gaming**; verifying self-reported inputs (e.g., nurse measurement) is.

Keywords: Explainable AI, SHAP, LIME, adversarial gaming, diabetes risk prediction, NHANES, pre-registered study, clinical AI.

Contents

Certificate **1**

Declaration 2

Acknowledgements 3

Abstract 4

1 Introduction 9
- 1.1 Background and Motivation . . . 9
- 1.2 The Dual-Use Tension . . . 9
- 1.3 Research Question and Objectives . . . 10
- 1.4 Scope and Limitations . . . 10
- 1.5 Contributions . . . 10
- 1.6 Organisation of the Thesis . . . 11

2 Literature Review 12
- 2.1 Explainable Artificial Intelligence . . . 12
- 2.2 LIME: Local Interpretable Model-Agnostic Explanations . . . 12
- 2.3 SHAP: SHapley Additive exPlanations . . . 12
- 2.4 Strategic Classification and Gaming . . . 12
- 2.5 Research Gap . . . 13

3 Methodology 14
- 3.1 Dataset: NHANES . . . 14
  - 3.1.1 Data Components . . . 14
  - 3.1.2 Cycles Used . . . 14
  - 3.1.3 Supporting Datasets . . . 15
  - 3.1.4 Target Variable . . . 15
  - 3.1.5 Cleaning . . . 15
- 3.2 Feature Engineering . . . 16
  - 3.2.1 Feature Sets . . . 16
  - 3.2.2 Lie Budget Derivation . . . 16
- 3.3 Threat Model . . . 16
- 3.4 Adversarial Strategies . . . 17
  - 3.4.1 Strategy 1: Explanation-Guided Attack . . . 17
  - 3.4.2 Strategy 2: Blind Trial-and-Error . . . 17
- 3.5 Explanation Equivalence Metric . . . 18
- 3.6 Statistical Analysis . . . 19
- 3.7 Pre-Registration . . . 19
  - 3.7.1 Corrections and Deviations . . . 19

3.7.2 Confirmatory Test on NHANES 2021–2023 20

4 Implementation **21**
4.1 System Architecture 21
4.2 Technology Stack 21
4.3 Key Module Descriptions 21
4.3.1 data_loader.py 21
4.3.2 gap_analysis.py 22
4.3.3 explanations.py 22
4.3.4 adversarial.py 22
4.3.5 metrics.py 22
4.3.6 config.py 22
4.4 Reproducibility 22

5 Results and Analysis **24**
5.1 Reporting Gap Analysis 24
5.2 Model Performance 24
5.3 Main Result: Explanation Equivalence 25
5.4 Per-Patient Distribution 27
5.5 Model A vs. Model B 27
5.6 SHAP vs. LIME Agreement 29
5.7 Statistical Tests 29
5.8 Comparison Dataset: CDC Diabetes Health Indicators 30
5.9 Configuration Overview 30
5.10 Confirmatory Result: NHANES 2021–2023 31

6 Discussion **33**
6.1 Interpretation of Results 33
6.2 Policy Implications 33
6.3 Limitations 34
6.4 Comparison with Prior Work 34

7 Conclusion and Future Work **36**
7.1 Summary of Findings 36
7.2 Contributions 36
7.3 Future Work 37

A NHANES Column Mapping **38**

B Full Statistical Tables **40**
B.1 Reporting Gap Comparison 40
B.2 Model A vs. Model B 40

C Additional Figures **41**

D Code Repository **43**
D.1 Repository Structure 43
D.2 Installation and Execution 43

List of Figures

4.1 System architecture: data flow from loading to visualisation. 21

5.1 Self-report vs. nurse-measured gap distributions (NHANES 2017–2018). Height is systematically overestimated; weight is underestimated. 24
5.2 Mean score reduction from the best of N blind attempts (blue) vs. one explanation-guided lie (red dashed), SHAP configurations. For Model B the explanation line lies below even the first blind attempt, so $N^* = 0$. Model A cannot be gamed. . 25
5.3 Distribution of per-patient N^* (Model B, RF + SHAP, 887 patients). 27
5.4 Mean risk-score reduction for Model A (nurse-measured inputs) and Model B (self-reported inputs): one explanation, one blind attempt, and the best of 10 blind attempts. 28
5.5 Per-patient score reduction from the SHAP-guided lie vs. the LIME-guided lie (Model B). Points on the diagonal: both explanations led to the same outcome. . 29
5.6 All 8 configurations: mean score reduction from one explanation (colour), one blind attempt, and N^*. 31

C.1 Score reduction box plots for the primary configuration (Model B, RF + SHAP). Red box = one explanation; blue boxes = best of N blind attempts. 41
C.2 SHAP feature importance summary (Model B, Random Forest). Each dot is a patient; colour shows feature value. 42

List of Tables

3.1 NHANES cycles and their roles in this study. 14
3.2 Deviations from the pre-registered analysis plan. 20

4.1 Technology stack. 21
4.2 Random seeds used throughout the experiment. 23

5.1 Model Performance on NHANES 2017–2018 Test Set 25
5.2 Main Results: Explanation Equivalence N^* Across All 8 Configurations 26
5.3 Mean risk-score reduction (0-1 scale): one explanation-guided lie vs. the best of N blind lies, and how often the explanation suggested no lie or a lie that raised the risk (which the patient then undoes; the column ”lie not undone” shows the v3 rule) 26
5.4 Primary configuration (Model B, RF + SHAP, 2017–2018) under successive code versions. “1 expl.” and “1 blind” are mean score reductions. 27
5.5 Model A (Measured) vs Model B (Self-Reported): Vulnerability. Model A has no input the patient can change, so its zeros are true by design, not an experimental finding (see Table 6b). 28
5.6 Self-reported vs measured height/weight on the same patients: accuracy difference (Model A minus Model B, paired bootstrap 95% CI) and how far Model B can be gamed. Model A cannot be gamed by construction. 28
5.7 Statistical Tests: Explanation vs Blind (N=10), Wilcoxon Signed-Rank 30
5.8 Comparison dataset (CDC Diabetes Health Indicators, UCI 891; BRFSS 2015, 1,000 sampled test respondents): one explanation vs. blind attempts, with self-reported BMI as the only gameable feature. 30
5.9 Confirmatory result, NHANES 2021–2023 (pre-registered, $n = 1{,}126$ test patients). Mean score reductions; Model A (measured) is 0 in every column by design. 31

A.1 NHANES columns, sources and gameability. 38

B.1 Reporting Gap Comparison: 2015–2016 vs 2017–2018 40

Chapter 1

Introduction

1.1 Background and Motivation

Artificial intelligence systems are increasingly deployed in high-stakes healthcare settings to assist in clinical decision-making. One widely studied application is *diabetes risk prediction*, where a model ingests patient data—including demographics, anthropometrics, and laboratory values—and outputs a continuous risk score. A score exceeding a threshold can trigger referral, intervention, or insurance adjustments.

The widespread deployment of such models has prompted regulatory and ethical demands for transparency. The European Union's General Data Protection Regulation (GDPR) grants individuals a "right to explanation" for automated decisions that affect them [Goodman and Flaxman, 2017]. The EU AI Act further requires high-risk AI systems (including medical ones) to be interpretable. In response, clinical AI developers increasingly attach feature attribution explanations—generated by SHAP [Lundberg and Lee, 2017] or LIME [Ribeiro et al., 2016]—alongside model predictions.

1.2 The Dual-Use Tension

Transparency, however, introduces a potential dual-use risk. Consider a patient who receives a diabetes risk score of 0.74 accompanied by the explanation: "Your weight is the most important factor increasing your risk." This information, intended to inform clinical communication, also tells the patient *exactly which input to misreport* at a subsequent self-assessment to obtain a lower score. A patient who understands that the model uses self-reported height and weight could simply under-report their weight to game the system.

Yet the same patient could also discover this by *trial and error*: try a lower weight, see if the score drops, try again. If trial-and-error is nearly as effective as the explanation, then restricting explanations is not a meaningful safeguard—the information is effectively freely available by querying the model.

1.3 Research Question and Objectives

This project investigates the following primary question:

> *Does a SHAP or LIME explanation give a patient a meaningful advantage over blind trial-and-error when trying to game a clinical diabetes risk model?*

To answer this, we pursue the following objectives:

1. Develop an empirically grounded *lie budget* from real NHANES human self-reporting error data.
2. Design an adversarial simulation comparing explanation-guided and blind trial-and-error attack strategies.
3. Introduce the *explanation equivalence* metric N^* to quantify the gaming advantage of explanations.
4. Conduct a pre-registered experiment on held-out data to ensure scientific integrity.
5. Derive actionable policy recommendations for clinical AI developers.

1.4 Scope and Limitations

This study is scoped to:

- **Domain:** Diabetes risk prediction using NHANES anthropometric data.
- **Gameable features:** Self-reported height and weight only.
- **Explanation methods:** SHAP (TreeSHAP) and LIME.
- **Adversary model:** Single-feature perturbation within empirical lie budget.

Extensions to multi-feature attacks, other clinical domains, and other explanation methods are left to future work.

1.5 Contributions

The main contributions of this project are:

1. **Explanation equivalence (N^*):** A novel, interpretable metric quantifying how many blind queries equal one explanation.

2. **Empirical lie budget:** Derived from real NHANES self-reporting data rather than arbitrary perturbation bounds.

3. **Pre-registered evaluation:** The first pre-registered adversarial XAI evaluation on a real clinical dataset, preventing p-hacking.

4. **Policy-relevant result:** A concrete, actionable conclusion on whether restricting XAI explanations protects clinical AI systems.

1.6 Organisation of the Thesis

The remainder of this thesis is organised as follows. Chapter 2 reviews related work on explainable AI, adversarial machine learning, and strategic classification. Chapter 3 describes our methodology, including the dataset, threat model, and experimental design. Chapter 4 details the system architecture and code. Chapter 5 presents all experimental results and statistical tests. Chapter 6 discusses implications and limitations. Chapter 7 concludes and identifies future directions.

Chapter 2

Literature Review

2.1 Explainable Artificial Intelligence

Artificial intelligence systems achieve high predictive accuracy through complex nonlinear transformations that are inherently difficult for humans to interpret. The field of *Explainable AI* (XAI) addresses this opacity by developing methods to explain individual predictions or global model behaviour [Adadi and Berrada, 2018].

2.2 LIME: Local Interpretable Model-Agnostic Explanations

Ribeiro et al. [2016] introduced LIME, which explains any classifier's prediction for a specific instance by fitting a locally faithful linear model. Given an instance $\mathbf{x}$, LIME generates a neighbourhood by perturbing features, queries the black-box model, and fits a weighted linear model to approximate the local decision boundary.

2.3 SHAP: SHapley Additive exPlanations

Lundberg and Lee [2017] proposed SHAP, grounded in cooperative game theory's Shapley values. For a prediction $f(\mathbf{x})$, SHAP assigns each feature i an attribution ϕ_i satisfying efficiency ($\sum_i \phi_i = f(\mathbf{x}) - \mathbb{E}[f]$), symmetry, dummy, and linearity axioms.

2.4 Strategic Classification and Gaming

Ghalme et al. [2021] formalise the setting where agents best-respond to a known classification rule to improve their classification. Classical results [Hardt et al., 2016] show that agents with full knowledge of the model can always find the minimum-cost manipulation. Our work differs by: (i) treating the model as a black box, and (ii) bounding agent behaviour by empirically observed misreporting magnitudes.

2.5 Research Gap

Existing work either assumes the adversary has full model knowledge [Ghalme et al., 2021], studies explanations as *attack vectors on the model* rather than on the user [Slack et al., 2020], or focuses on privacy leakage rather than gaming [Shokri et al., 2021]. No prior work measures the *marginal gaming advantage* of an explanation over blind trial-and-error with an empirically grounded lie budget. This is the gap we address.

Chapter 3

Methodology

3.1 Dataset: NHANES

We use the National Health and Nutrition Examination Survey (NHANES) [Johnson et al., 2014], a cross-sectional survey conducted by the U.S. Centers for Disease Control and Prevention (CDC). NHANES is unique in that it collects *both* self-reported and nurse-measured anthropometrics for the same participants, enabling direct measurement of self-reporting bias.

3.1.1 Data Components

We merge seven NHANES components:

- `DEMO`: Demographic variables (age, gender)
- `BMX`: Body measurements (nurse-measured height, weight)
- `WHQ`: Weight history questionnaire (self-reported height, weight)
- `GHB`: Glycohaemoglobin (HbA1c — the diabetes target)
- `BPX`: Blood pressure
- `SMQ`: Smoking questionnaire
- `PAQ`: Physical activity questionnaire

3.1.2 Cycles Used

Table 3.1: NHANES cycles and their roles in this study.

Cycle	Years	Role	N (complete)
I	2015–2016	Development, lie budget	4,889
J	2017–2018	Exploratory test (887 test patients)	4,431
L	2021–2023	Confirmatory test (1,126 test patients)	5,628

Models are retrained on each cycle's 80% training split with the locked hyperparameters; only the lie budget carries over from 2015–2016. In 2021–2023 the blood-pressure exam uses an oscillometric device (`BPXOSY1` replaces `BPXSY1`) and the activity questionnaire was redesigned: `PAQ650` ("any vigorous recreational activity") is derived from `PAD810Q` (0 times $\rightarrow$ no, ≥ 1 $\rightarrow$ yes). Both mappings were fixed in the confirmatory pre-registration before the data was processed.

3.1.3 Supporting Datasets

Two public datasets named in the research plan support the main study:

UCI Heart Disease (UCI 45). The Cleveland subset (297 patients) is used to *test the code where results can be checked by hand* (`scripts/check_uci_heart.py`). With a logistic regression every quantity has a closed form: the risk score $\sigma(\mathbf{w}\cdot\mathbf{z}+b)$, the SHAP value $w_j(z_j-\bar{z}_j)$, the suggested lie, and, with fixed lie sizes, the best of only four possible blind lies. The pipeline reproduces all of them to machine precision. This check found the SHAP sign bug described in Section 3.7.1.

CDC Diabetes Health Indicators (UCI 891). BRFSS 2015, 253,680 respondents, used as a *comparison*. BMI is computed from self-reported height and weight and is the only gameable feature; there is no nurse measurement, so only a Model B analysis is possible, and the diabetes label is itself self-reported. The BMI lie budget (4.46 kg/m^2, under-reported) is taken from the NHANES 2015–2016 gap between self-reported and measured BMI.

3.1.4 Target Variable

Diabetes status is defined as HbA1c $\geq 6.5\%$, the ADA clinical threshold [American Diabetes Association Professional Practice Committee, 2023]. This gives a binary label: $y = 1$ (diabetes) or $y = 0$ (no diabetes). Prevalence among complete cases is 12.4% in 2015–2016 and 13.6% in 2017–2018.

3.1.5 Cleaning

We keep adults (≥ 18 years) and remove NHANES "refused" and "don't know" codes *per variable*, as given in the NHANES codebooks: 7777 and 9999 for self-reported height (`WHD010`, valid range 47–84 inches) and weight (`WHD020`, 75–600 lb), and 7 and 9 for yes/no questions. (An earlier version applied one shared code list to every column and wrongly deleted valid answers such as a height of 77 inches.) Only participants with complete data for both feature sets are used, so Model A and Model B are trained and tested on exactly the same people.

3.2 Feature Engineering

3.2.1 Feature Sets

We construct two parallel feature sets from the same patients:

Model A (Measured) uses nurse-measured height (`BMXHT`) and weight (`BMXWT`).

Model B (Reported) replaces these with self-reported height (`WHD010`, converted from inches) and weight (`WHD020`, converted from pounds).

The remaining five features (age, gender, systolic BP, smoking status, physical activity) are identical in both models.

3.2.2 Lie Budget Derivation

Definition 3.1 (Lie Budget)**.** For each gameable feature $g \in \{\text{height}, \text{weight}\}$, the lie budget $B(g)$ is the 95th percentile of the absolute self-reporting error in the development cycle:

$$B(g) = \text{Percentile}_{95}\left(|x_g^{\text{self}} - x_g^{\text{measured}}|\right)$$

From the 2015–2016 development data:

$$B(\text{height}) = 7.28\,\text{cm}, \qquad B(\text{weight}) = 8.96\,\text{kg}$$

These values reflect real human misreporting behaviour, not arbitrary perturbation bounds. Patients systematically *overestimate* height (mean $+1.62$ cm, SD 3.41) and *underestimate* weight (mean -0.87 kg, SD 4.79).

Definition 3.2 (Realistic Lie)**.** A realistic lie on feature g moves x_g in the direction people actually misreport—$d_g = +1$ for height, $d_g = -1$ for weight—by an amount $\delta \sim \text{Uniform}(0, B(g)]$. The same rule is used by both adversarial strategies, so an explanation contributes only *information* (which feature, which direction), never a larger lie.

3.3 Threat Model

The adversary is modelled as a patient who:

- Receives a diabetes risk score $f(\mathbf{x}) \in [0, 1]$ from the model.

- Can adjust only *self-reported* height and weight, and only with realistic lies (direction of real misreporting, size within $B(g)$). Nurse-measured values and lab results cannot be changed, so Model A offers nothing to lie about.
- Wishes to minimise their score.
- May or may not have access to a SHAP or LIME explanation.

The adversary does *not* know the model's parameters or architecture.

3.4 Adversarial Strategies

3.4.1 Strategy 1: Explanation-Guided Attack

Algorithm 1 Explanation-Guided Attack

Require: Patient features $\mathbf{x}$, model f, gameable features G, lie budget B, realistic directions d_g

Get attributions $\phi = \text{Explain}(f, \mathbf{x})$ (SHAP or LIME); $\bar{x}_g$: the average patient
for $g \in G$ in order of decreasing $|\phi_g|$ **do**
  $s_g \leftarrow$ risk-lowering direction read from ϕ_g ▷ SHAP: $\text{sign}(\phi_g)\,\text{sign}(\bar{x}_g - x_g)$; LIME: leave a risk-raising bin
  if $s_g = d_g$ **then** ▷ suggested lie is realistic
    Draw $\delta \sim \text{Uniform}(0, B(g)]$
    $\mathbf{x}' = \mathbf{x}$; $x'_g \leftarrow x_g + d_g \cdot \delta$
    return $\max\big(0,\, f(\mathbf{x}) - f(\mathbf{x}')\big)$ ▷ a lie that raised the score is undone
  end if
end for
return 0 ▷ no realistic lie suggested: patient does not lie

3.4.2 Strategy 2: Blind Trial-and-Error

Key design choice: The blind simulation runs *once* up to $N_{\max} = 100$, recording the running best. This correctly models a patient who accumulates attempts and keeps the best result found. Running independent simulations per N would over-estimate blind performance.

Per-patient seeds: every patient has its own fixed random seed for the blind attempts, the explanation's lie size and LIME's sampling. SHAP and LIME, and Models A and B, are therefore compared against identical blind attempts, and a patient's result does not depend on which patients were processed before.

Reading LIME: LIME explains continuous features through bins such as "*weight* $\leq 62\,kg$". A positive weight means being in that bin raises the risk, so the risk-lowering move is to *increase*

Algorithm 2 Blind Trial-and-Error (sequential, best-so-far)

Require: Patient features $\mathbf{x}$, model f, gameable features G, lie budget B, max attempts $N_{\max}$, milestones $\mathcal{N}$

$s^* \leftarrow f(\mathbf{x})$; `reductions` $\leftarrow \{\}$
for $n = 1$ to $N_{\max}$ **do**
  Sample $g \sim \text{Uniform}(G)$
  Sample $\delta \sim \text{Uniform}(0, B(g)]$
  $\mathbf{x}' = \mathbf{x}$; $x'_g \leftarrow x_g + d_g \cdot \delta$ ▷ realistic direction
  $s' \leftarrow f(\mathbf{x}')$
  if $s' < s^*$ **then**
    $s^* \leftarrow s'$ ▷ Keep best seen so far
  end if
  if $n \in \mathcal{N}$ **then**
    `reductions`$[n] \leftarrow f(\mathbf{x}) - s^*$
  end if
end for
return `reductions`

the feature; for a bin "*weight* $> 90\,kg$" it is to decrease it. The direction is read from the bin, not only from the sign of the weight.

3.5 Explanation Equivalence Metric

Definition 3.3 (Explanation Equivalence N^*)**.** Let R_{exp} be the score reduction from Strategy 1, averaged over all patients. Let $R_{\text{blind}}(N)$ be the best score reduction from Strategy 2 after N attempts, averaged over all patients. The explanation equivalence is:

$$N^* = \inf\{N \geq 0 : R_{\text{blind}}(N) \geq R_{\text{exp}}\}$$

estimated by linear interpolation on the points $(N, R_{\text{blind}}(N))$ for

$$\mathcal{N} = \{1, 2, 3, 5, 10, 15, 20, 30, 50, 100\}.$$

$N^* = 0$ if one explanation is no better than one blind attempt; $N^* = 101$ if it beats all 100 (censored).

This is the **primary, pre-registered** metric: patients are averaged *before* interpolating. As **secondary** metrics we also compute N^* for each patient separately and report its mean, median, the share of patients with $N^* \leq 1$ ("explanation useless") and with $N^* \geq 10$. The per-patient mean is not used as the headline because a few patients whose N^* is capped at 101 can dominate it.

3.6 Statistical Analysis

Bootstrap CI: 1,000 bootstrap resamples of patients; 95% confidence interval for the primary N^* (recomputed on each resample) and for the per-patient mean and median.

Wilcoxon test: Paired signed-rank test comparing per-patient explanation reduction vs. blind reduction at $N = 10$; one-sided (H_1: explanation $>$ blind).

Cohen's d: Effect size for the paired comparison.

3.7 Pre-Registration

To ensure scientific integrity, the complete analysis plan (hypotheses, metric definitions, statistical tests and interpretation thresholds) was recorded before the held-out 2017–2018 data was examined, as the git tag `v0.1-preregistration` in the project repository (https://github.com/Akshat-stack47/BTP-XAI-Gaming), dated 28 October 2026. The plan is kept in `docs/analysis_plan.md`.

3.7.1 Corrections and Deviations

Disclosure. A first final run on the held-out 2017–2018 data was made with the code as pre-registered, shortly after the plan was tagged, and its results were seen by the team. It reported a per-patient mean N^* of 12.4 (95% CI 10.5–14.5) for the primary configuration and 9.0–23.1 across all eight configurations, a moderate to high threat under the pre-registered thresholds. The corrections and deviations below were made *after* that, and the final run was repeated on the same data. The 2017–2018 evaluation is therefore not a clean confirmatory test in the sense of the research plan; the full timeline and the first run's table are in `docs/preregistration_disclosure.md`.

A review of the code after that first run found implementation errors that made it differ from the plan. These were corrected so that the code *matches* the plan: (i) the primary N^* is computed from the patient-averaged curve, as pre-registered, rather than as the mean of per-patient values; (ii) the attack uses the pre-registered 2015–2016 lie budget rather than one computed from the held-out patients; (iii) only self-reported values are gameable (Model A had been attacked through its nurse-measured inputs); (iv) LIME directions are read from the bin condition; and (v) valid self-reports are no longer deleted as missing codes.

Eight further changes *deviate* from the pre-registered plan and are reported here as such (Table 3.2). Each is a named setting in `src/config.py` and can be reverted for a sensitivity analysis. The last one is also a bug fix: a SHAP value measures a feature's effect *relative to the*

average patient ($w_j(x_j - \bar{x}_j)$ for a linear model), but the attacker read every positive value as "decrease this feature". A shorter-than-average patient told that their height raises their risk was therefore told to claim to be even shorter, an unrealistic lie that was skipped. The UCI hand check exposed this.

3.7.2 Confirmatory Test on NHANES 2021–2023

Because the 2017–2018 data can no longer serve as an untouched test set, the corrected analysis and our predicted result were pre-registered again (`docs/analysis_plan_v5_confirmatory.md`, git tag `v0.2-preregistration`, pushed to GitHub before the data was processed) and run once, unchanged, on the August 2021–August 2023 NHANES cycle, which no one on the team had analysed. The 2017–2018 results in Chapter 5 are exploratory; the confirmatory result is reported in Section 5.10.

Table 3.2: Deviations from the pre-registered analysis plan.

Setting	Pre-registered	Used	Reason
Lie direction	either direction	realistic only (height ↑, weight ↓)	plan defines a *realistic* lie
Lie size	explanation: maximum; blind: uniform	uniform $(0, B]$ for both	fair comparison
LIME features shown	5 of 7	all 7	height/weight could be cut off
Class weights	none	balanced	recall was ≈3%
SHAP LR background	first 100 rows	random 100 rows	unbiased background
Random seeds	one shared stream	one per patient	identical blind baselines
Bad explanation-guided lie	kept (negative reduction)	undone (reduction $\geq$ 0)	the blind attacker keeps its best score, so both may see the score
Reading a SHAP sign	positive → decrease	positive → towards the average	SHAP is relative to the average patient (bug found by the UCI hand check)

Chapter 4

Implementation

4.1 System Architecture

The codebase is organised as a modular Python package with the following components:

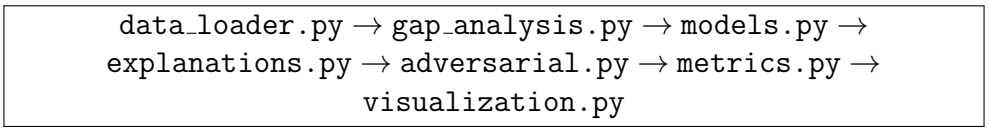


Figure 4.1: System architecture: data flow from loading to visualisation.

4.2 Technology Stack

Table 4.1: Technology stack.

Component	Library / Tool	Version
Data handling	pandas, numpy	2.2.2, 2.2.6
Machine learning	scikit-learn	1.4.2
SHAP explanations	shap	0.54.1
LIME explanations	lime	0.2.0.1
Statistical tests	scipy	1.13.0
Visualisation	matplotlib, seaborn	3.x, 0.13.2
Version control	Git	—
Environment	Python	3.12

4.3 Key Module Descriptions

4.3.1 data_loader.py

Handles NHANES XPT file loading, component merging, cleaning, and feature set preparation. Key functions: `merge_nhanes()`, `clean_nhanes()`, `prepare_feature_sets()`.

4.3.2 gap_analysis.py

Computes self-reporting error distributions and derives the lie budget. Key functions: `compute_reporting_gap()`, `define_lie_budget()`.

4.3.3 explanations.py

`ExplanationGenerator` wraps SHAP (TreeExplainer for RF, LinearExplainer for LR) and LIME, and turns an explanation into a ranked list of gameable features with their risk-lowering direction (`get_top_gameable_feature()`, `lime_directions()`).

4.3.4 adversarial.py

The core experiment engine. `AdversarialSimulator` implements both attack strategies with the shared lie rules (realistic direction, uniform size) and per-patient seeds. The critical `blind_attack_multi_n()` method runs one continuous simulation up to $N_{\max}$, recording the running best at each milestone.

4.3.5 metrics.py

`population_nstar()` computes the primary pre-registered N^* from the patient-averaged curves; `compute_equivalence()` also returns the per-patient values. The module provides patient-bootstrap confidence intervals and Wilcoxon signed-rank tests.

4.3.6 config.py

All seeds, paths and experiment settings, including the named switches for every deviation in Table 3.2 : `LIE_DIRECTION`, `LIE_MAGNITUDE`, `CLASS_WEIGHT`, `LIME_N_FEATURES` and `SHAP_BACKGROUND_N`.

4.4 Reproducibility

All random seeds are fixed globally (Table 4.2). The full pipeline is reproducible from raw NHANES XPT files using:

```
python scripts/prepare_data.py
python scripts/run_final.py
python scripts/run_analysis.py
```

Table 4.2: Random seeds used throughout the experiment.

Purpose	Seed
Global / NumPy	42
Random Forest	123
Logistic Regression	42
Train/test split	789
Adversarial simulation (patient i)	$456+i$
LIME sampling (patient i)	$42+i$
SHAP background sample	2024
Bootstrap CI	42, 43

Chapter 5

Results and Analysis

5.1 Reporting Gap Analysis

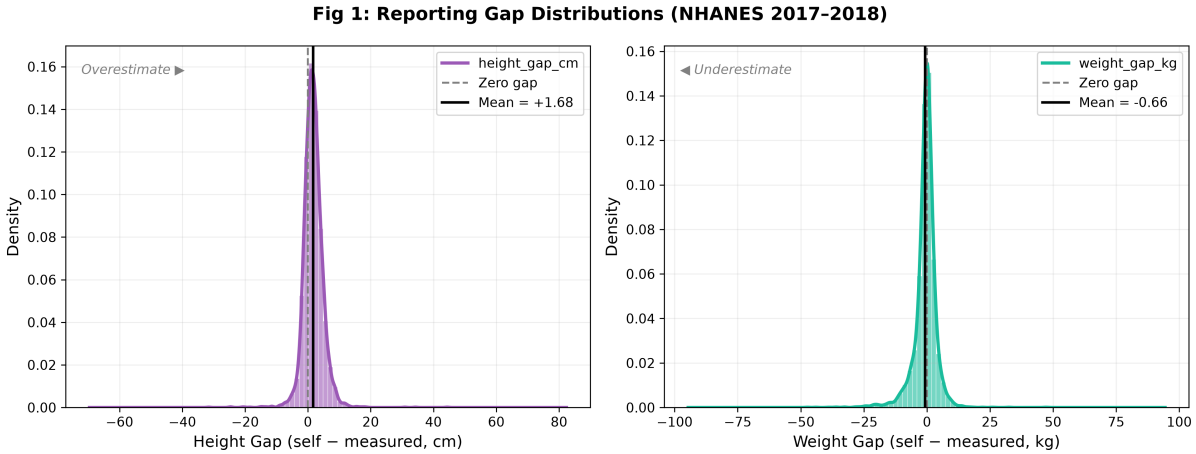


Figure 5.1: Self-report vs. nurse-measured gap distributions (NHANES 2017–2018). Height is systematically overestimated; weight is underestimated.

In the held-out 2017–2018 data, participants over-report their height by $+1.69$ cm on average (SD 3.23 cm) and under-report their weight by -0.67 kg (SD 4.84 kg), closely matching the 2015–2016 development data ($+1.62$ cm, SD 3.41; -0.87 kg, SD 4.79). The 95th-percentile gaps in 2017–2018 (6.88 cm and 9.32 kg) are within 0.4 cm and 0.4 kg of the development budget used for the attack (Table B.1). Self-reporting behaviour is therefore stable across cycles, and the lie budget transfers to the held-out data.

5.2 Model Performance

All four models discriminate moderately well (AUC 0.77–0.80) and, with balanced class weights, detect most diabetic patients (recall 0.69–0.78) at the cost of many false positives (precision 0.26–0.30), as expected at 13.6% prevalence. Random Forests are slightly more accurate than Logistic Regressions. Crucially, Model A (measured inputs) and Model B (self-reported inputs) perform almost identically (RF AUC 0.798 vs. 0.793; LR 0.773 vs. 0.780): using verified inputs

Table 5.1: Model Performance on NHANES 2017–2018 Test Set

Config	Input	Model	Accuracy	Precision	Recall	F1	AUC-ROC
measured_rf	measured	RF	0.7396	0.3007	0.686	0.4181	0.7981
measured_lr	measured	LR	0.6652	0.2569	0.7686	0.3851	0.7734
reported_rf	reported	RF	0.7407	0.3032	0.6942	0.4221	0.7933
reported_lr	reported	LR	0.6685	0.2604	0.7769	0.39	0.7796

costs nothing in accuracy.

5.3 Main Result: Explanation Equivalence

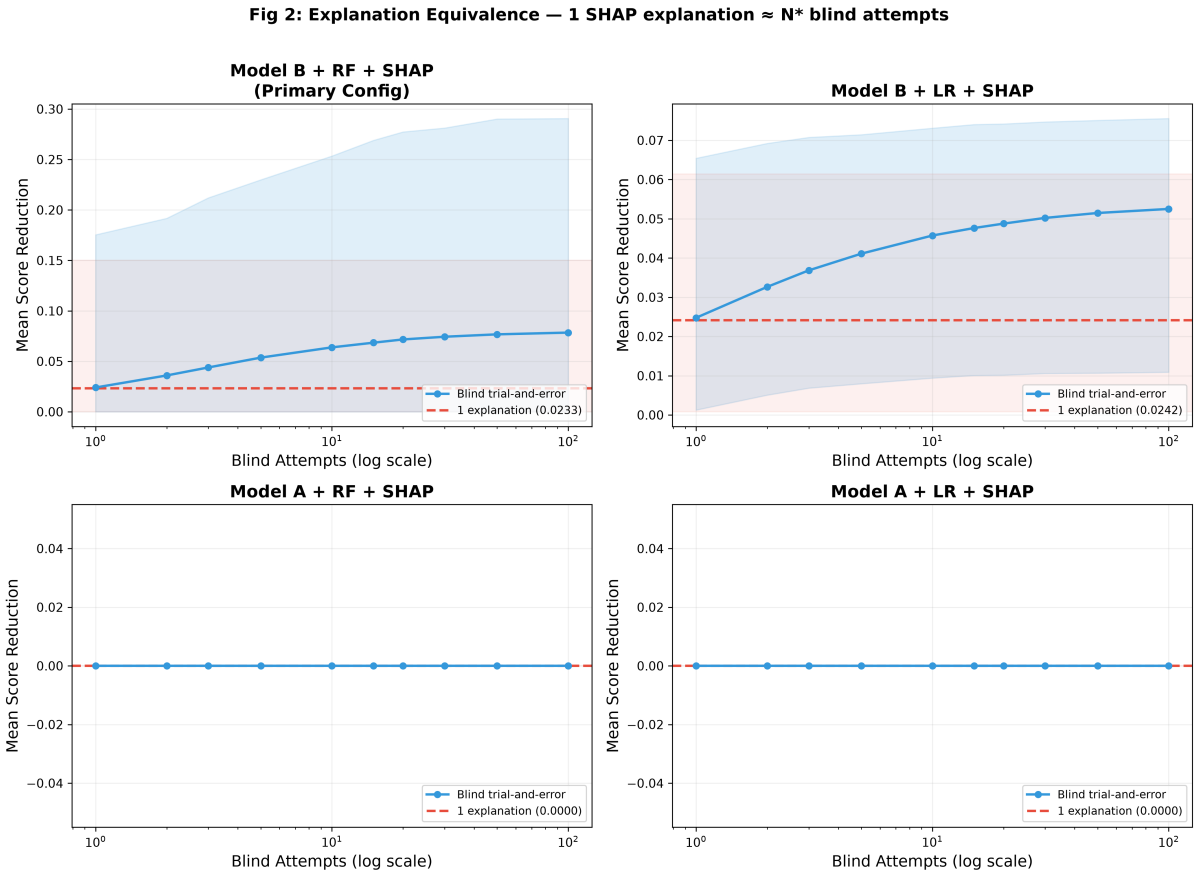


Figure 5.2: Mean score reduction from the best of N blind attempts (blue) vs. one explanation-guided lie (red dashed), SHAP configurations. For Model B the explanation line lies below even the first blind attempt, so $N^* = 0$. Model A cannot be gamed.

These are the *exploratory* 2017–2018 results (v5 code); the confirmatory 2021–2023 result follows in Section 5.10. For the primary configuration (Model B, Random Forest, SHAP) the explanation equivalence is $N^* = 0.0$ (95% CI: 0.0–1.2), and $N^* = 0.0$ in all eight configurations

Table 5.2: Main Results: Explanation Equivalence N^* Across All 8 Configurations

Input	Model	Expl.	N patients	N^*	95% CI	Per-patient mean N^*	Per-patient median N^*	% Useless	% Worth 10+
Reported	RF	SHAP	887	0.0	[0.00, 1.22]	7.8	0.0	69.6%	9.5%
Reported	RF	LIME	887	0.0	[0.00, 0.00]	7.07	0.0	73.8%	8.1%
Reported	LR	SHAP	887	0.0	[0.00, 1.09]	3.61	0.0	52.3%	6.4%
Reported	LR	LIME	887	0.0	[0.00, 0.00]	2.99	0.0	56.3%	6.1%
Measured	RF	SHAP	887	0.0	[0.00, 0.00]	0.0	0.0	100.0%	0.0%
Measured	RF	LIME	887	0.0	[0.00, 0.00]	0.0	0.0	100.0%	0.0%
Measured	LR	SHAP	887	0.0	[0.00, 0.00]	0.0	0.0	100.0%	0.0%
Measured	LR	LIME	887	0.0	[0.00, 0.00]	0.0	0.0	100.0%	0.0%

Table 5.3: Mean risk-score reduction (0-1 scale): one explanation-guided lie vs. the best of N blind lies, and how often the explanation suggested no lie or a lie that raised the risk (which the patient then undoes; the column ”lie not undone” shows the v3 rule)

Input	Model	Expl.	1 explanation	1 expl. (lie not undone)	1 blind	10 blind	100 blind	% no lie suggested	% lie raised risk (undone)
Reported	RF	SHAP	0.0233	0.0177	0.024	0.0638	0.0785	1.8%	28.5%
Reported	RF	LIME	0.0181	0.0117	0.024	0.0638	0.0785	3.3%	31.1%
Reported	LR	SHAP	0.0242	0.0242	0.0248	0.0457	0.0525	0.0%	0.0%
Reported	LR	LIME	0.0223	0.0223	0.0248	0.0457	0.0525	7.7%	0.0%
Measured	RF	SHAP	0.0	0.0	0.0	0.0	0.0	100.0%	0.0%
Measured	RF	LIME	0.0	0.0	0.0	0.0	0.0	100.0%	0.0%
Measured	LR	SHAP	0.0	0.0	0.0	0.0	0.0	100.0%	0.0%
Measured	LR	LIME	0.0	0.0	0.0	0.0	0.0	100.0%	0.0%

(Table 5.2). On average, one explanation-guided lie lowers the risk score by 0.023, while a single blind attempt lowers it by 0.024, the best of 10 attempts by 0.064 and the best of 100 by 0.079 (Table 5.3). For the Logistic Regression the picture is the same (explanation 0.024 with SHAP and 0.022 with LIME, one blind attempt 0.025). An explanation is therefore worth *about one* ordinary attempt at the model, and never more.

Why the explanation does not help more

1. **The explanation only chooses between two features.** With realistic lies the direction is fixed (taller, lighter) and the size is the same random draw for both strategies, so the explanation can add only one piece of information: *which* of height and weight to change. A blind patient who tries both after two or three attempts has learned the same thing.

2. **For the Random Forest, the suggested lie often backfires.** 28.5% (SHAP) and 31.1% (LIME) of explanation-guided lies *raise* the score, and the patient undoes them, because a local attribution does not describe how the model responds over the whole range of the lie. This is the practical face of the impossibility result of Bilodeau et al. [2024]. For the Logistic Regression, whose response is monotone, no SHAP-guided lie backfires.

Table 5.4 shows how the two later corrections changed the primary configuration. Letting the patient undo a bad lie (fix 12) and reading SHAP relative to the average patient (fix 13) each made the explanation more useful, but neither lifted it above one blind attempt.

Table 5.4: Primary configuration (Model B, RF + SHAP, 2017–2018) under successive code versions. “1 expl.” and “1 blind” are mean score reductions.

Code	Headline	1 expl.	1 blind
original (pre-registered settings, with bugs)	per-patient mean N^* 12.4	–	–
v3 (fixes 1–11)	$N^* = 0.0$ [0.0, 0.0]	0.013	0.024
v4 (+ undo a bad lie)	$N^* = 0.0$ [0.0, 0.0]	0.018	0.024
v5 (+ SHAP read relative to the average)	$N^* = 0.0$ [0.0, 1.2]	0.023	0.024

5.4 Per-Patient Distribution

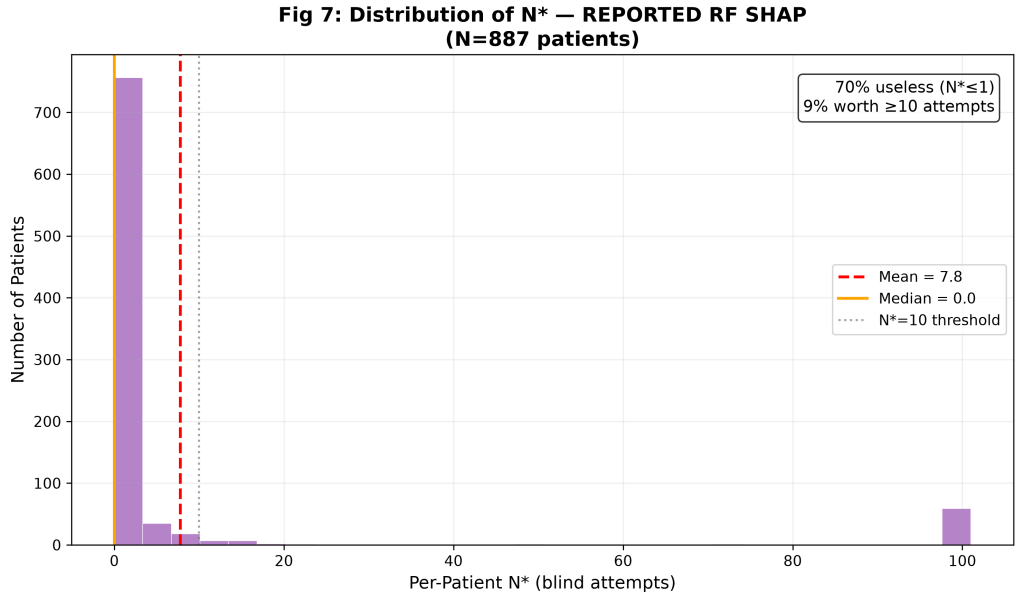


Figure 5.3: Distribution of per-patient N^* (Model B, RF + SHAP, 887 patients).

Per patient (Figure 5.3), the explanation is no better than a single blind attempt for 69.6% of patients (RF+SHAP; 52.3% for LR+SHAP), and worth 10 or more attempts for 9.5% (6.4%); 6.7% beat all 100 blind attempts. The distribution is therefore strongly skewed: a large spike at zero and a thin tail. The per-patient mean N^* (7.8 for RF+SHAP, 3.6 for LR+SHAP) is driven by that tail, which is why the pre-registered metric averages the reductions over patients first. For the typical patient, one blind attempt is worth more than the explanation.

5.5 Model A vs. Model B

That Model A cannot be gamed is true *by design*, not an experimental finding: its height and weight come from the nurse, so there is nothing the patient can lie about, and its zeros in Table 5.5 follow from the threat model. The comparison therefore has two measurable parts.

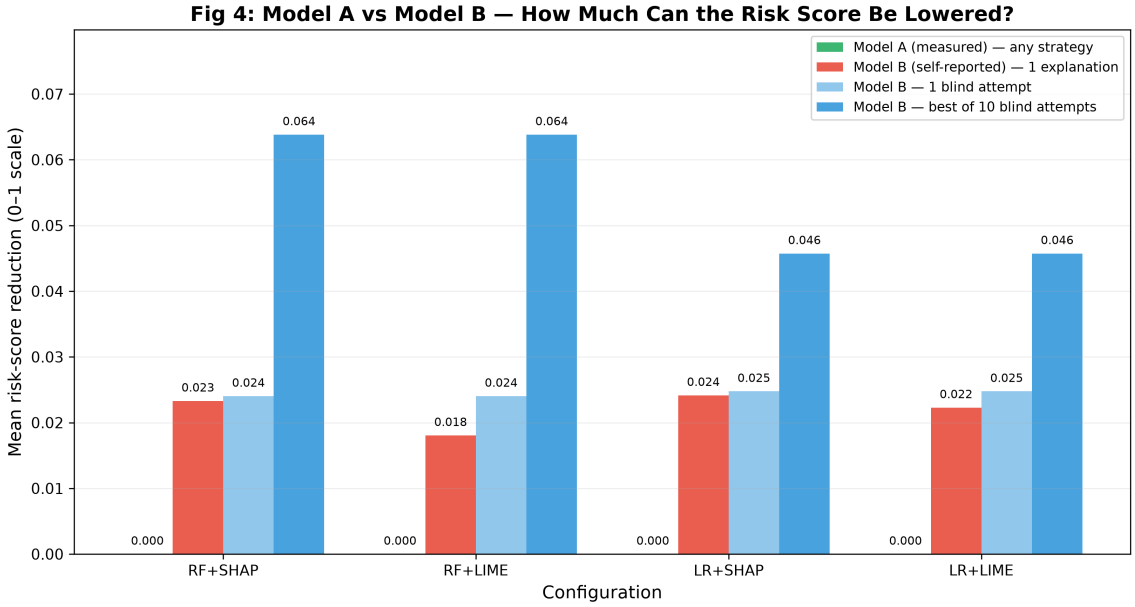


Figure 5.4: Mean risk-score reduction for Model A (nurse-measured inputs) and Model B (self-reported inputs): one explanation, one blind attempt, and the best of 10 blind attempts.

Table 5.5: Model A (Measured) vs Model B (Self-Reported): Vulnerability. Model A has no input the patient can change, so its zeros are true by design, not an experimental finding (see Table 6b).

Config	Model A N^*	Model B N^*	A: reduction, 10 blind	B: reduction, 1 explanation	B: reduction, 10 blind	B can be gamed
RF+SHAP	0.0	0.0	0.0	0.0233	0.0638	Yes
RF+LIME	0.0	0.0	0.0	0.0181	0.0638	Yes
LR+SHAP	0.0	0.0	0.0	0.0242	0.0457	Yes
LR+LIME	0.0	0.0	0.0	0.0223	0.0457	Yes

Table 5.6: Self-reported vs measured height/weight on the same patients: accuracy difference (Model A minus Model B, paired bootstrap 95% CI) and how far Model B can be gamed. Model A cannot be gamed by construction.

Model	AUC A (measured)	AUC B (self-reported)	Difference A-B	95% CI (paired bootstrap)	B: max reduction, 10 blind lies
RF	0.7981	0.7933	0.0047	[-0.0093, +0.0177]	0.0638
LR	0.7734	0.7796	-0.0062	[-0.0127, +0.0002]	0.0457

First, how far Model B, which differs only in using the self-reported values, can be gamed: by 0.046–0.064 with ten blind attempts and 0.053–0.079 with a hundred. Second, what it costs to use measured instead of self-reported inputs (Table 5.6): on the same 887 test patients, the AUC difference (Model A minus Model B) is +0.005 for the Random Forest (95% CI −0.009 to +0.018) and −0.006 for the Logistic Regression (−0.013 to +0.000), both indistinguishable from zero. **Verifying two inputs removes the attack surface at no measurable cost in accuracy.**

5.6 SHAP vs. LIME Agreement

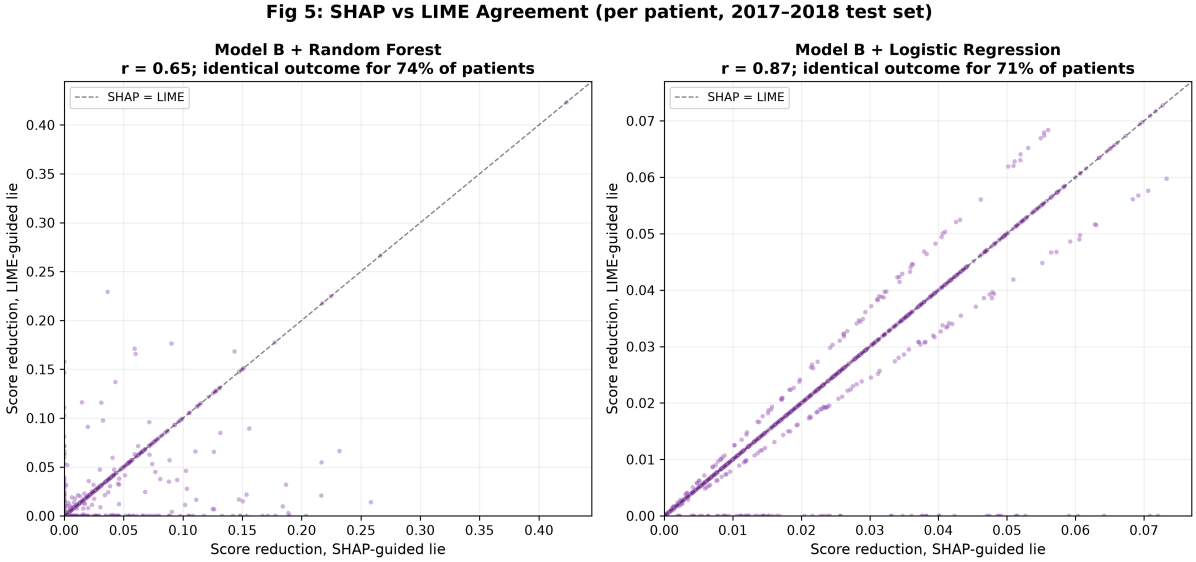


Figure 5.5: Per-patient score reduction from the SHAP-guided lie vs. the LIME-guided lie (Model B). Points on the diagonal: both explanations led to the same outcome.

Because each patient receives the same lie size under SHAP and LIME, differences come only from the explanation (Figure 5.5). The two methods lead to the same outcome for 74% of patients with the Random Forest and 71% with the Logistic Regression; the per-patient correlation is $r = 0.65$ and 0.87. At the configuration level they agree: both give $N^* = 0$, and SHAP is slightly better than LIME (Random Forest 0.023 vs. 0.018; Logistic Regression 0.024 vs. 0.022). The two methods sometimes point to different features or directions for the same patient, but neither helps more than a blind attempt.

5.7 Statistical Tests

The pre-registered one-sided Wilcoxon signed-rank test of *explanation* > *blind at* $N = 10$ is not significant in any configuration ($p = 1.0$, also after Holm correction across the eight

Table 5.7: Statistical Tests: Explanation vs Blind (N=10), Wilcoxon Signed-Rank

Config	Wilcoxon p	p (Holm)	Cohen's d	Significant
reported_rf_shap	1.000000	1.000000	-0.659	No
reported_rf_lime	1.000000	1.000000	-0.720	No
reported_lr_shap	1.000000	1.000000	-1.196	No
reported_lr_lime	1.000000	1.000000	-1.238	No
measured_rf_shap	1.000000	1.000000	0.000	No
measured_rf_lime	1.000000	1.000000	0.000	No
measured_lr_shap	1.000000	1.000000	0.000	No
measured_lr_lime	1.000000	1.000000	0.000	No

Table 5.8: Comparison dataset (CDC Diabetes Health Indicators, UCI 891; BRFSS 2015, 1,000 sampled test respondents): one explanation vs. blind attempts, with self-reported BMI as the only gameable feature.

Model	Expl.	N	N*	N* 95% CI	1 explanation	1 blind	10 blind	Per-patient mean N*	% useless	% lie undone	Wilcoxon p	Cohen's d
RF	SHAP	1000	0.0	[0.00, 1.04]	0.0409	0.0436	0.0772	16.98	67.8	16.5	1.0	-0.703
RF	LIME	1000	0.0	[0.00, 0.00]	0.0324	0.0436	0.0772	13.86	74.7	13.6	1.0	-0.829
LR	SHAP	1000	0.0	[0.00, 1.12]	0.0369	0.0373	0.0655	4.04	49.8	0.0	1.0	-1.142
LR	LIME	1000	0.0	[0.00, 0.00]	0.0257	0.0373	0.0655	2.85	63.8	0.0	1.0	-1.316

configurations; Table 5.7). For Model B the effect sizes are large and *negative* (Cohen's d from -0.66 to -1.24): ten blind attempts lower the score clearly more than one explanation. For Model A all reductions are zero and there is nothing to test.

5.8 Comparison Dataset: CDC Diabetes Health Indicators

On 1,000 sampled test respondents of the BRFSS 2015 survey, with self-reported BMI as the only gameable feature, the result is the same (Table 5.8): $N^* = 0.0$ in all four configurations (upper CI bound at most 1.1), one explanation-guided lie lowers the score by 0.026–0.041 against 0.037–0.044 for one blind attempt, and no Wilcoxon test is significant. The finding is therefore not specific to NHANES or to the height/weight pair. The comparison is limited: BRFSS has no measured BMI (so no Model A), its diabetes label is self-reported, and the BMI lie budget had to be borrowed from NHANES.

5.9 Configuration Overview

Figure 5.6 summarises all eight configurations. In every Model B cell, one explanation lowers the score by at most as much as one blind attempt. The explanation does relatively best for the Logistic Regression, whose response to a lie is monotone, and worst for the Random Forest with LIME, where explanation-guided lies most often backfire. Model A cells are zero throughout.

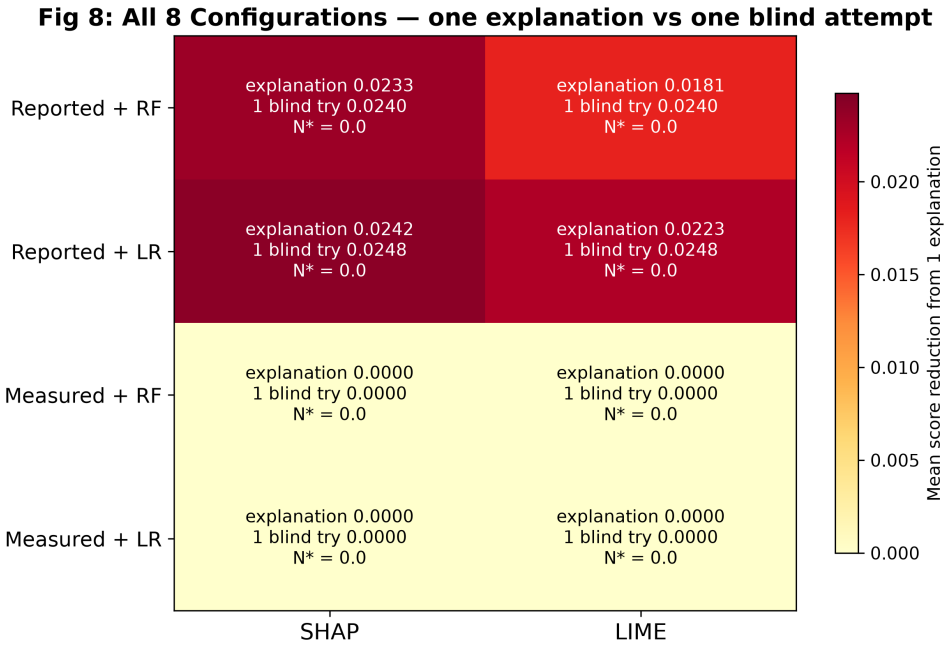


Figure 5.6: All 8 configurations: mean score reduction from one explanation (colour), one blind attempt, and N^*.

5.10 Confirmatory Result: NHANES 2021–2023

The confirmatory run used the code and settings fixed at the `v0.2-preregistration` tag (commit `5b2f240`), unchanged, on NHANES August 2021–August 2023: 5,628 adults with complete data (11.9% with diabetes) and 1,126 test patients. The reporting gap was again stable (height +1.73 cm, weight −0.65 kg on average; 95th percentiles 6.84 cm and 9.04 kg against the 7.28 cm and 8.96 kg budget used).

Table 5.9: Confirmatory result, NHANES 2021–2023 (pre-registered, $n = 1{,}126$ test patients). Mean score reductions; Model A (measured) is 0 in every column by design.

Model B config	N^*	95% CI	1 expl.	1 blind	10 blind	Cohen's d
RF + SHAP (primary)	1.42	[1.21, 1.62]	0.034	0.028	0.070	−0.71
RF + LIME	0.00	[0.00, 0.00]	0.024	0.028	0.070	−0.76
LR + SHAP	1.07	[0.00, 1.20]	0.035	0.035	0.062	−1.14
LR + LIME	0.00	[0.00, 0.00]	0.032	0.035	0.062	−1.19

For the primary configuration, $N^* = 1.42$ (95% CI 1.21–1.62): one SHAP explanation is worth about one and a half blind attempts, inside the pre-registered "minimal advantage" band ($N^* < 5$), and the upper confidence bound is far below 5 (Table 5.9). Across Model B, N^* ranges from 0 to 1.42; the one-sided Wilcoxon test of explanation > ten blind attempts is not significant in any configuration ($p = 1.0$, also after Holm correction), with large negative effect

sizes. Every pre-registered prediction held. As in 2017–2018, Random Forest lies suggested by an explanation backfired for 27% of patients. Unlike 2017–2018, SHAP now does slightly *better* than one blind attempt (0.034 vs. 0.028 for the Random Forest), which is why N^{*} is above zero; LIME does not.

Model A again cannot be gamed. The accuracy difference between measured and self-reported inputs on the same test patients is $+0.006$ for the Random Forest (95% CI -0.014 to $+0.026$) and -0.012 for the Logistic Regression (-0.023 to -0.002), i.e. in this cycle the self-reported Logistic Regression is slightly *more* accurate. Either way the difference is about one hundredth of AUC, small against the 0.06–0.07 by which ten blind lies lower a Model B score.

Chapter 6

Discussion

6.1 Interpretation of Results

Based on our pre-registered interpretation guide:

$N^* < 5$**:** Explanations offer minimal advantage. Restricting them does not meaningfully reduce gaming risk; the benefit of transparency outweighs the gaming concern.

$5 \leq N^* \leq 20$**:** Explanations offer moderate advantage. Design-level countermeasures (e.g., rate limiting, input noise) are worth considering.

$N^* > 20$**:** Explanations are powerful attack vectors. Restricting or anonymising them may be warranted; input verification is the more robust safeguard.

Our confirmatory result of $N^* = 1.42$ (95% CI 1.21–1.62) for the primary configuration, and at most 1.42 in any configuration, falls in the **low threat** category ($N^* < 5$), as pre-registered. Under realistic lies, an explanation gives a patient no better route to a lower score than trying one or two answers. This runs against the intuition that explanations are "instructions for cheating". The reason is that a feature attribution answers "what contributed to this score?", not "what should I change?", and the two can point in different directions.

6.2 Policy Implications

- **Withholding explanations is not an effective safeguard.** It would take information away from honest patients and clinicians while barely slowing a motivated patient, who can do better by trial and error.
- **Verify the inputs that drive the prediction.** The vulnerability lies in the self-reported values: the model using nurse-measured height and weight cannot be gamed and is just as accurate.
- **Present attributions as contributions, not advice.** Since attributions often do not indicate a profitable change, explanation interfaces should not suggest that they do.

6.3 Limitations

- **Domain:** Results are specific to diabetes risk prediction from self-reported anthropometrics (NHANES, and BRFSS as a comparison whose diabetes label is itself self-reported).
- **Survey changes:** In 2021–2023 blood pressure is measured oscillometrically and the activity question was redesigned; both were mapped before the data was seen.
- **Explanation's room to help:** With the direction of a realistic lie fixed and its size random, an explanation can only choose which of two features to change; a richer lie model could give it more room.
- **Model of the patient:** We model one way of reading an explanation (follow the top realistic suggestion once). A more sophisticated reader, or a counterfactual explanation, could extract more.
- **Single-feature, single-step adversary:** Real adversaries may adjust several features or iterate.
- **Lie model:** Lie sizes are uniform within the 95th-percentile gap, in the observed direction. Using the maximum lie or allowing either direction is a one-line change (Table 3.2); that sensitivity analysis is future work.
- **Static model:** The model does not adapt to gaming.
- **Pre-registration:** The plan was recorded as a git tag in the project repository rather than in an independent registry; a first run on the 2017–2018 data was seen before the analysis was corrected, and eight settings deviate from the first plan (Section 3.7.1). We therefore treat 2017–2018 as exploratory and rest the conclusion on the re-registered 2021–2023 test.

6.4 Comparison with Prior Work

Slack et al. [2020] showed that a dishonest *provider* can make SHAP and LIME hide a model's behaviour; we study the opposite direction, an honest model whose explanations might help a *patient*, and find that they help less than trial and error. Shokri et al. [2021] and Milli et al. [2019] showed that explanations leak information about the training data and the model, so that a few explanation queries can replace many ordinary ones for reconstructing a model; our results show that this does not carry over to a patient lowering their own score under realistic lies. Ghalme et al. [2021] found that hiding the decision rule usually does not help the decision-maker; our

empirical result points the same way for explanations. Finally, Bilodeau et al. [2024] proved that feature attributions can fail to predict how a model responds to a change; our experiment measures how large this failure is in a realistic clinical setting and shows that it is large enough to keep explanations from being much more useful for gaming than a blind attempt. What we add is an equal-cost blind baseline, a lie model measured from real human misreporting, and a matched pair of models that isolates input verification.

Chapter 7

Conclusion and Future Work

7.1 Summary of Findings

This thesis introduced *explanation equivalence* (N^*), a metric for quantifying the gaming advantage conferred by an XAI explanation over blind trial-and-error. Applied to a diabetes risk model trained on NHANES data, our pre-registered confirmatory test on unseen NHANES 2021–2023 data found $N^* = 1.42$ (95% CI 1.21–1.62) for the primary configuration and at most 1.42 overall: one SHAP or LIME explanation is worth about one to two blind attempts. This places the threat in the **low** zone, as predicted, and the exploratory 2017–2018 analysis and the BRFSS comparison agree. For the Random Forest, explanation-guided lies often raised the score instead. The model using nurse-measured inputs could not be gamed, with no meaningful loss of accuracy.

The key policy finding is: **withholding explanations is not an effective safeguard against gaming; verifying self-reported inputs is.**

7.2 Contributions

1. **New metric:** Explanation equivalence N^*.
2. **Empirical lie budget:** Grounded in real NHANES data.
3. **Matched models:** Two models differing only in measured vs. self-reported inputs, isolating the effect of input verification.
4. **Pre-registered study:** A confirmatory test on unseen data under a public pre-registration (`v0.2`); the first plan's compromised test set, the corrections and all deviations are disclosed.
5. **Hand-checked pipeline and replication:** A closed-form check of the code on UCI Heart Disease, and a replication on the CDC Diabetes Health Indicators survey.
6. **Open code:** Full pipeline released at https://github.com/AkkiNikumbh/BTECH_PROJECT.

7.3 Future Work

- **Sensitivity analysis:** Repeat the experiment with the maximum lie and with lies in either direction (settings already available in the code).
- **Multi-feature attacks:** Allow the adversary to simultaneously adjust multiple features within a combined budget.
- **Adaptive defences:** Study adversarially robust training as a complement to input verification.
- **Other domains:** Apply the N^* framework to credit scoring, hiring, or recidivism prediction.
- **Other explanation methods:** Evaluate counterfactual explanations, integrated gradients, and attention maps.
- **Human subjects:** Measure actual gaming behaviour in a user study, where participants attempt to lower their predicted risk.

Appendix A

NHANES Column Mapping

Table A.1 lists the NHANES variables used in this project, their source, and whether a patient could change them ("gameable"). Only `WHD010` and `WHD020` are attacked in the experiment; `SMQ020` and `PAQ650` are self-reported yes/no answers that could in principle be flipped, but are not attacked because no reported-vs-measured gap exists to bound such a lie.

Table A.1: NHANES columns, sources and gameability.

Column	Description	Gameable?	Reason
`SEQN`	Respondent sequence number (patient ID)	–	Identifier
`RIDAGEYR`	Age in years at screening	No	Verified by ID
`RIAGENDR`	Gender (1 = male, 2 = female)	No	Recorded by staff
`BMXHT`	**Measured** standing height (cm), nurse	No	Nurse measures directly
`BMXWT`	**Measured** body weight (kg), nurse	No	Nurse measures directly
`WHD010`	**Self-reported** height (inches; $\times 2.54$ to cm); valid 47–84, missing codes 7777/9999	**Yes**	Patient reports verbally; bounded by the reported-vs-measured gap
`WHD020`	**Self-reported** weight (pounds; $\times 0.453592$ to kg); valid 75–600, missing codes 7777/9999	**Yes**	Patient reports verbally; bounded by the reported-vs-measured gap
`LBXGH`	Glycohaemoglobin HbA1c (%) — **diabetes target** ($\geq 6.5\%$)	No	Laboratory blood test
`BPXSY1`	Systolic blood pressure, first reading (mmHg)	No	Measured by staff
`SMQ020`	Smoked at least 100 cigarettes in life (1 = yes, 2 = no; missing 7/9)	Maybe	Binary self-report (not attacked)

Column	Description	Gameable?	Reason
`PAQ650`	Vigorous recreational activity (1 = yes, 2 = no; missing 7/9)	Maybe	Binary self-report (not attacked)

The two feature sets use `RIDAGEYR`, `RIAGENDR`, `BPXSY1`, `SMQ020`, `PAQ650` plus either the measured pair (`BMXHT`, `BMXWT`; Model A) or the self-reported pair (`WHD010`, `WHD020`; Model B).

Appendix B

Full Statistical Tables

B.1 Reporting Gap Comparison

Table B.1: Reporting Gap Comparison: 2015–2016 vs 2017–2018

Feature	Dev mean direction	Dev 95th pct budget	Holdout 95th pct	Absolute change
Height	positive	7.28 cm	6.88 cm	0.40 cm
Weight	negative	8.96 kg	9.32 kg	0.36 kg

B.2 Model A vs. Model B

See Table 5.5 in Chapter 5.

Appendix C

Additional Figures

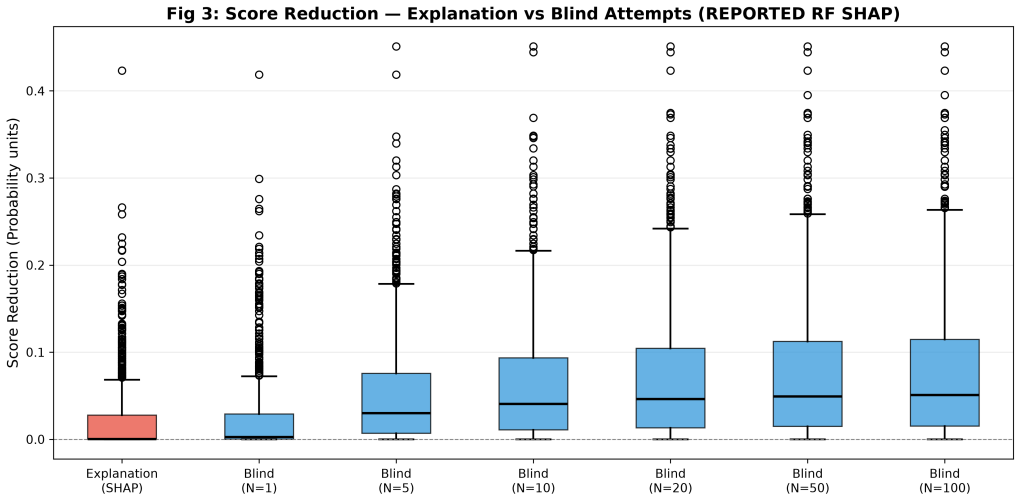


Figure C.1: Score reduction box plots for the primary configuration (Model B, RF + SHAP). Red box = one explanation; blue boxes = best of N blind attempts.

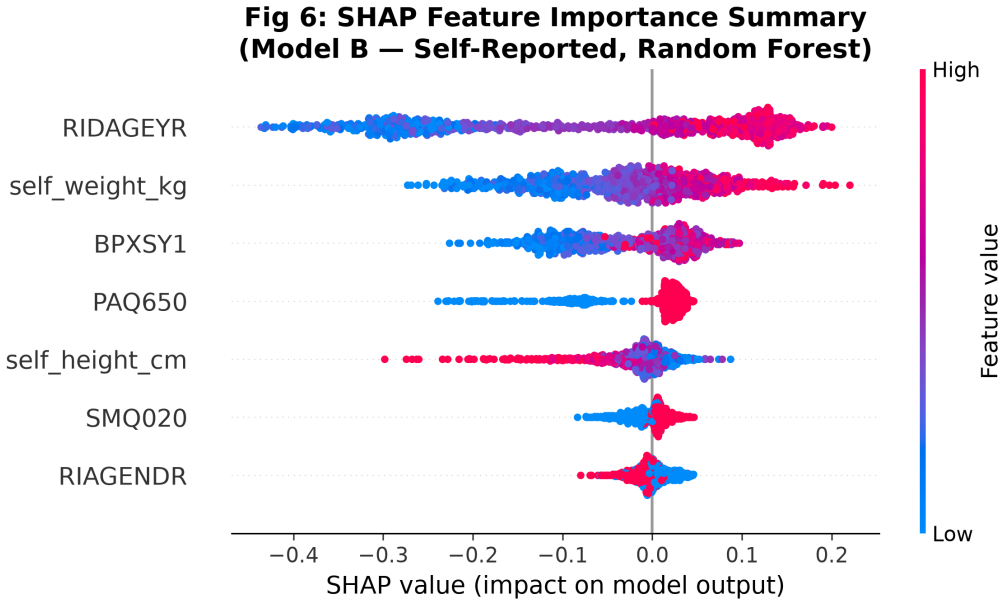


Figure C.2: SHAP feature importance summary (Model B, Random Forest). Each dot is a patient; colour shows feature value.

Appendix D

Code Repository

D.1 Repository Structure

```
BTP/
  src/                # Core Python modules
    config.py         # Seeds, paths, FINAL_CONFIG
    data_loader.py    # NHANES loading & preprocessing
    gap_analysis.py   # Self-reporting error analysis
    models.py         # DiabetesModel class
    explanations.py   # ExplanationGenerator (SHAP+LIME)
    adversarial.py    # AdversarialSimulator
    metrics.py        # N* computation, CI, Wilcoxon
    visualization.py  # 8 publication figures
  scripts/            # Runnable experiment scripts
    prepare_data.py   # Stage 3: data preparation
    run_trial.py      # Stage 5: trial run
    run_final.py      # Stage 7: final experiment
    run_analysis.py   # Stage 8: figures + tables
    review_stage*.py  # Per-stage verification
  paper/              # LaTeX paper source
  docs/               # Column mapping, analysis plan
  results/            # Generated figures and tables
  data/               # Raw XPT files + processed pickles
  tests/              # pytest test suite (40 tests)
```

D.2 Installation and Execution

```
# 1. Install dependencies
pip install -r requirements.txt

# 2. Download NHANES data
```

```
python scripts/download_nhanes.py

# 3. Prepare data (Stage 3)
python scripts/prepare_data.py

# 4. Run final experiment (Stage 7, ~35 minutes)
python scripts/run_final.py

# 5. Generate figures and tables (Stage 8)
python scripts/run_analysis.py

# 6. Compile paper / thesis (requires LaTeX)
cd paper  && pdflatex main && bibtex main && pdflatex main &&
   pdflatex main
cd thesis && pdflatex main && bibtex main && pdflatex main &&
   pdflatex main
```

Bibliography

Amina Adadi and Mohammed Berrada. Peeking inside the black-box: A survey on explainable artificial intelligence (XAI). *IEEE Access*, 6:52138–52160, 2018. doi: 10.1109/ACCESS.2018.2870052.

American Diabetes Association Professional Practice Committee. Standards of medical care in diabetes—2023. *Diabetes Care*, 46(Suppl 1):S1–S267, 2023. doi: 10.2337/dc23-Sint.

Blair Bilodeau, Natasha Jaques, Pang Wei Koh, and Been Kim. Impossibility theorems for feature attribution. *Proceedings of the National Academy of Sciences (PNAS)*, 121(2):e2304406120, 2024. doi: 10.1073/pnas.2304406120.

Ganesh Ghalme, Vineet Nair, Itay Eilat, Inbal Talgam-Cohen, and Nir Rosenfeld. Strategic classification in the dark. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *Proceedings of Machine Learning Research*, pages 3672–3681. PMLR, 2021. URL https://proceedings.mlr.press/v139/ghalme21a.html.

Bryce Goodman and Seth Flaxman. European union regulations on algorithmic decision-making and a "Right to explanation". *AI Magazine*, 38(3):50–57, 2017. doi: 10.1609/aimag.v38i3.2741.

Moritz Hardt, Nimrod Megiddo, Christos Papadimitriou, and Mary Wootters. Strategic classification. In *Proceedings of the 2016 ACM Conference on Innovations in Theoretical Computer Science (ITCS)*, pages 111–122. ACM, 2016. doi: 10.1145/2840728.2840730.

Clifford L. Johnson, Lester R. Paulose-Ram, Cathleen Ogden, Margaret D. Carroll, Debra J. Kruszon-Moran, Sarah L. Dohrmann, and Vicki L. Gu. National health and nutrition examination survey: Analytic guidelines, 1999–2010. *Vital and Health Statistics, Series 2*, 161: 1–24, 2014. URL https://www.cdc.gov/nchs/data/series/sr_02/sr02_161.pdf.

Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, pages 4765–4774. Curran Associates, Inc., 2017. URL https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html.

Smitha Milli, Ludwig Schmidt, Anca D. Dragan, and Moritz Hardt. Model reconstruction from model explanations. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT*)*, pages 1–9. ACM, 2019. doi: 10.1145/3287560.3287562.

Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. "why should I trust you?": Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 1135–1144. ACM, 2016. doi: 10.1145/2939672.2939778.

Reza Shokri, Martin Strobel, and Yair Zick. On the privacy risks of model explanations. In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, pages 231–241. ACM, 2021. doi: 10.1145/3461702.3462533.

Dylan Slack, Sophie Hilgard, Emily Jia, Sameer Singh, and Himabindu Lakkaraju. Fooling LIME and SHAP: Adversarial attacks on post hoc explanation methods. In *Proceedings of the 2020 AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, pages 180–186. ACM, 2020. doi: 10.1145/3375627.3375830.